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Batch No: 2CS56

Panel No:38

Group No:3

20CS295 – Mini Project I

"Development of AI-Driven Social Platform with FAQ Support "

Type of Project: Application

Expected Project Outcome : Product

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Introduction

- In many colleges, event updates and announcements are conveyed manually via students/class Representatives.
- Social media is not specifically built for college needs—thus missing relevant academic features.....
- There is no dedicated platform that provides access to past year question papers and academic materials in an organized way. Our chatbot is enhanced to instantly share links to past year question papers on request.
- Some institutions use Learning Management Systems (LMS) like Moodle or Google Classroom, but they do not integrate chatbot-based FAQs, social interaction, or student analytics in one platform.
- Our project addresses these gaps through a unified web platform with Firebase, AI-powered FAQ Chatbot, and NLP-based text analyzer.

Problem identification, scope and objectives (Contd.,)

Problem Definition

- Students lack a **dedicated, secure, and interactive college platform** for communication, event updates, academic resources, and FAQs. Existing solutions (social media groups, notice boards) are fragmented, unreliable, and not tailored for college needs.

Scope

A secure, college-only platform for SREC where students can post content, view club events, access past papers, and get real-time help through an AI chatbot — all in one place to boost academic and social engagement.

Problem identification, scope and objectives

Objectives

- Enable students to post content, view club events, and connect socially, with a sentiment analyzer to understand post tone.
- Build a secure, centralized platform for all SREC students — especially newcomers — to access academic and campus information easily.
- Provide quick access to past year question papers and real-time help through an AI-powered FAQ chatbot.

Literature Survey - 01

Experiment	Inference	Problem gap
Development of a conversational AI bot prototype designed to support efficient learning by providing intelligent, interactive, and personalized assistance to learners.	The study showed that the AI bot boosts learning by increasing engagement, giving timely guidance, and supporting adaptive pathways, proving its value as an educational tool.	For further work, it is suggested to use different geometries with the modification methods that is associated with the increasing evaporation surface area such as fins and cylinders.

- **Source:** Ahmed Elragal, Ali Ismail Awad, Ingemar Andersson, and Jörgen Nilsson, "A Conversational AI Bot for Efficient Learning: A Prototypical Design", IEEE Access, vol. 12, 2024, DOI: 10.1109/ACCESS.2024.3476953.

Literature Survey - 02

Experiment	Inference	Problem gap
Developed and deployed MoodleBot, an LLM-based chatbot using a Retrieval-Augmented Generation (RAG) architecture integrated into Moodle LMS, and evaluated it using the Technology Acceptance Model (TAM) and accuracy metrics.	Achieved high perceived usefulness and ease of use, with 88% accuracy in answering course-related queries, effectively supporting self-regulated learning.	The solution remains effective but faces issues like hallucinations, bias, scalability, and missing a social-style platform for student interaction, which CampusConnect addresses.

- **Source:** Neumann, A. T., Yin, Y., Sowe, S., Decker, S., & Jarke, M. (2025). An LLM-Driven Chatbot in Higher Education for Databases and Information Systems. *IEEE Trans. Educ.*, 68(1). https://doi.org/10.1109/TE.2024.3467912

Literature Survey - 03

Experiment	Inference	Problem gap
Analyzed student participation in online courses using engagement data collected across multiple semesters. Applied time-series forecasting models (ARIMA and Prophet) to identify temporal patterns and critical factors influencing academic retention.	Found that participation is highest at the beginning of courses but declines over time. Peer interaction, course length, and engagement strategies were key factors affecting student retention and performance in online education.	AI models found participation patterns, but lacked an interactive system for engagement, real-time support, and peer interaction—gaps filled by CampusConnect's AI chatbot and community features.

- **Source:** William Villegas-Ch, Joselin García-Ortiz, and Santiago Sánchez-Viteri, "Application of Artificial Intelligence in Online Education: Influence of Student Participation on Academic Retention in Virtual Courses", IEEE Access, May 2024, DOI: 10.1109/ACCESS.2024.3403758.

Literature Survey - 04

Experiment	Inference	Problem gap
Studied how chatbot-human personality match affects behavior, with 54 students using chatbots of three personalities; measured engagement, trust, and use intention via surveys and interviews.	Findings showed personality congruence boosted trust and engagement, especially for extroverted users, while effects were minimal for conscientious and agreeable participants.	Personality congruence boosted trust but was limited to small, fixed settings; CampusConnect fills this gap with dynamic, personalized chatbot responses for lasting engagement.

- **Source:** Mohammad Amin Kuhail, Mohamed Bahja, Ons Al-Shamaileh, Justin Thomas, Amina Alkazemi, and Joao Negreiros, "Assessing the Impact of Chatbot-Human Personality Congruence on User Behavior: A Chatbot-Based Advising System Case", IEEE Access, May 2024, DOI: 10.1109/ACCESS.2024.3402977.

Literature Survey - 05

Experiment	Inference	Problem gap
Applied NLP with Bag of Words + Logistic Regression on the Dreaddit dataset to detect stress in Reddit academic posts, achieving 77.78% accuracy (F1=0.79) and 72% on manual validation.	The study found ~29% of academic subreddit posts show stress, highest in professors' communities, highlighting ML's value for early stress detection in online discussions.	The model was limited to Reddit text and simple validation, while CampusConnect adds AI chatbots for real-time stress detection and support.

- **Source:** Nazzere Oryngozha, Pakizar Shamoi, and Ayan Igali, "Detection and Analysis of Stress-Related Posts in Reddit's Academic Communities", IEEE Access, Jan 2024, DOI: 10.1109/ACCESS.2024.3357662.

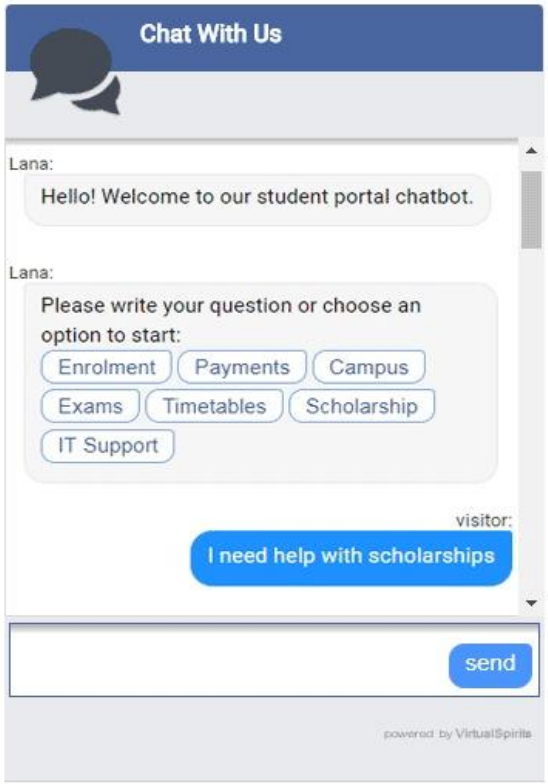
Existing technology of the project

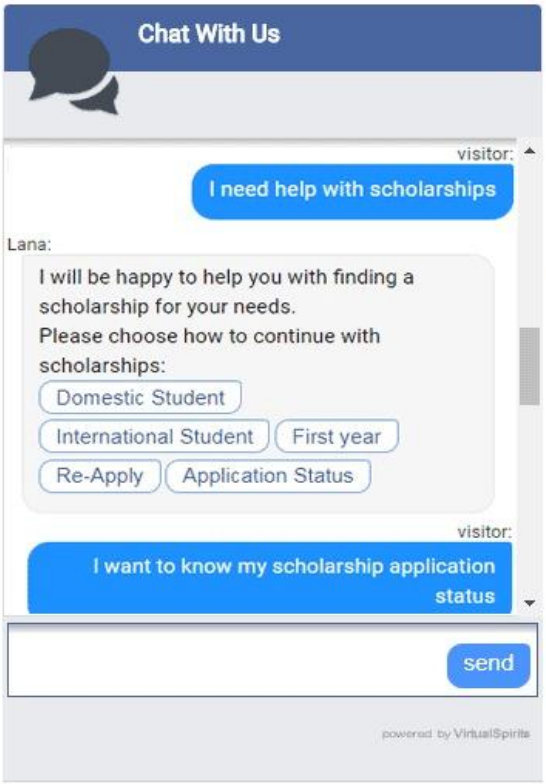
- Personality-Adapted Chatbots: Chatbots that adjust their tone, language, and responses based on user personality traits to create better engagement.
- Educational Advising Bots: Chatbots used as virtual academic advisors that can respond to student queries about curriculum, scheduling, and institutional policies.
- User Behavior Analysis: Systems that evaluate how users interact with different chatbot personalities to improve UX and retention.
- Emotionally Congruent Interfaces: Development of chatbot interfaces and response patterns that feel more “human,” improving user satisfaction and interaction depth.

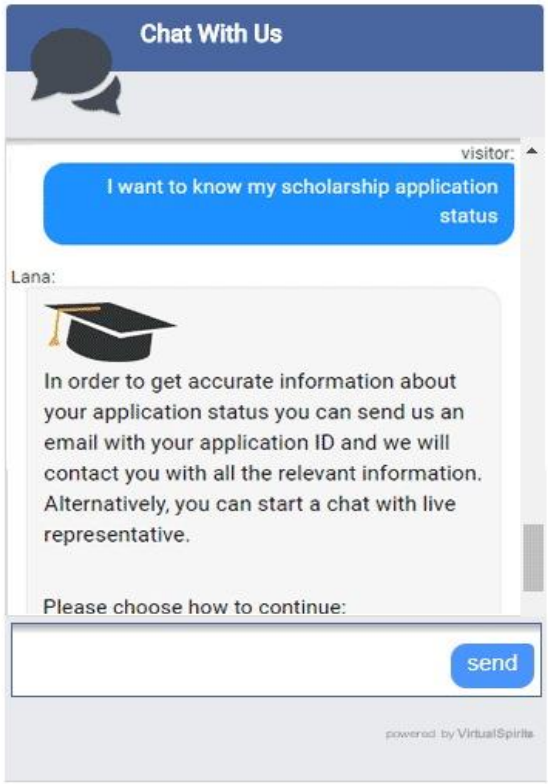
Source: “Assessing the Impact of Chatbot-Human Personality Congruence on User Behavior: A Chatbot-Based Advising System Case”, Mohammad Amin Kuhail, Mohamed Bahja, Ons Al-Shamaileh, Justin Thomas, Amina Alkazemi, and Joao Negreiros, IEEE Access, Vol. 12, No. –, pp. –, 2024. (Scopus Indexed)

Existing technology of the project

Photographs

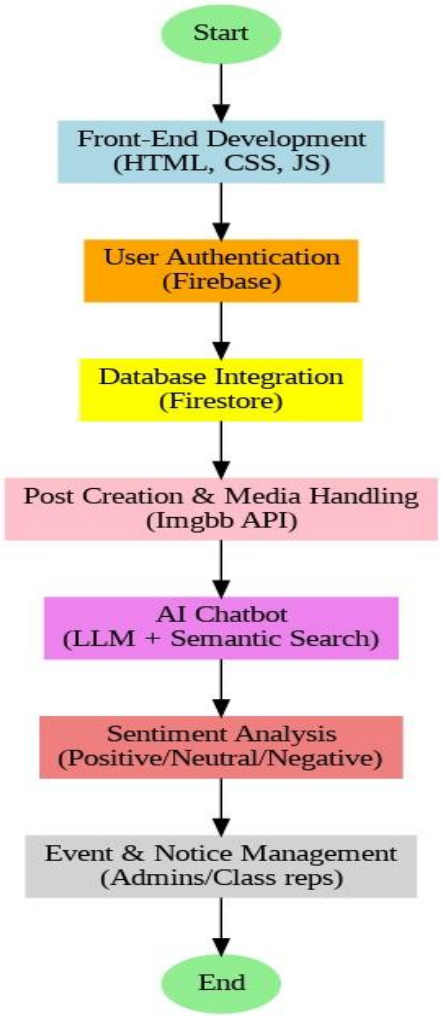




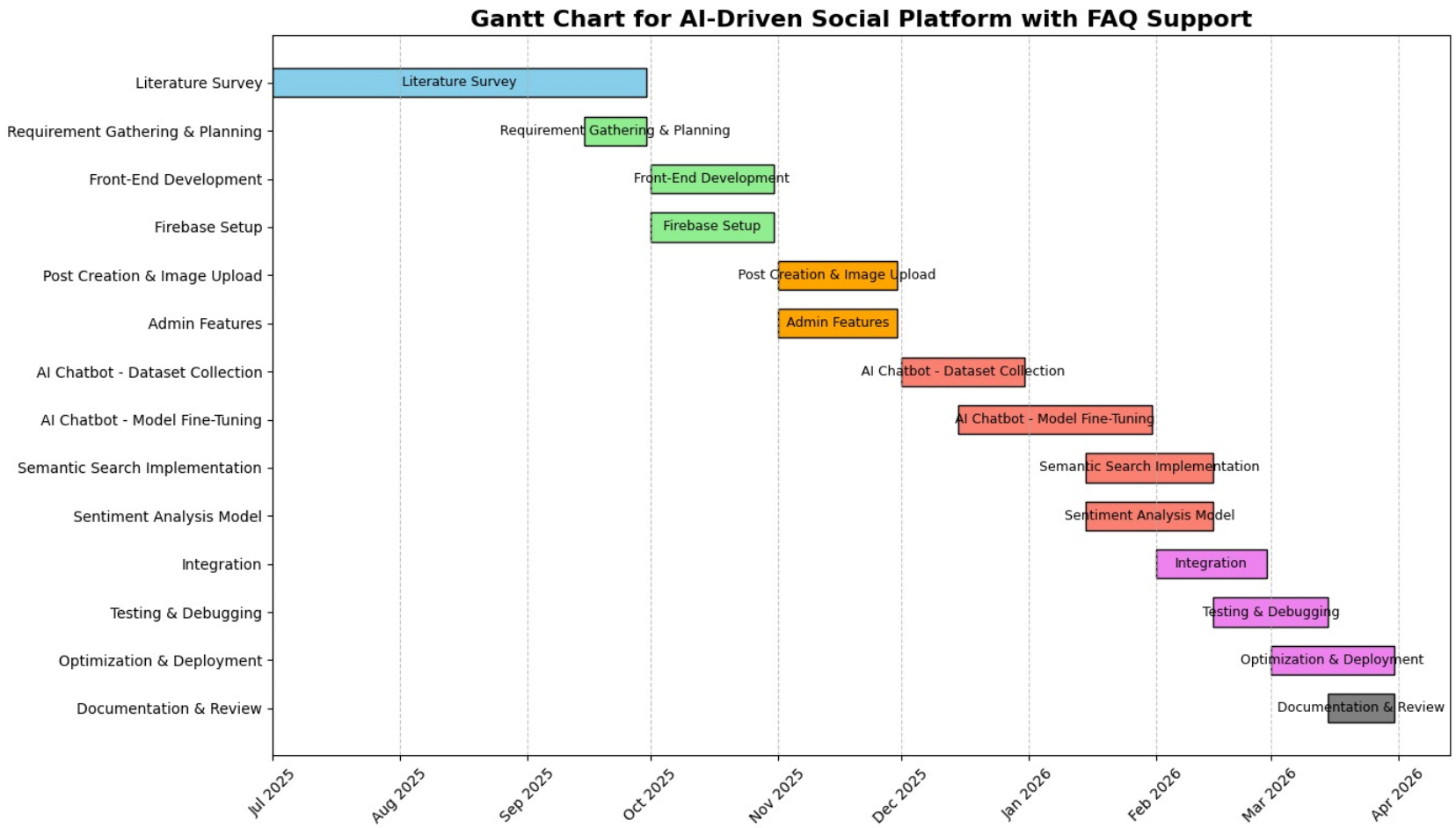


Methodology to be adopted

Flow Chart



Timeline of the Project and Budget



Proposed Outcome of the Project

- A user-friendly web-based college social networking platform exclusively for SREC students.
- Enables students to post thoughts, share images, and interact within a secure campus-only environment.
- Integrated AI-powered chatbot for answering frequently asked questions (FAQs), including:
- General college queries (fees, timings, departments, SREC website summary).
- Club-related queries and event information.
- Access to past-year question papers via chatbot & Google Drive links.
- Sentiment analyzer integrated with each post to assess and display post emotions (positive, neutral, negative).
- Future-ready design that can support:
- Club event announcements.

References

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